"...one of the most highly regarded and expertly designed C++ library projects in the world."
— Herb Sutter and Andrei Alexandrescu, C++ Coding Standards

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# Creation and Modification

## *Template parameters*

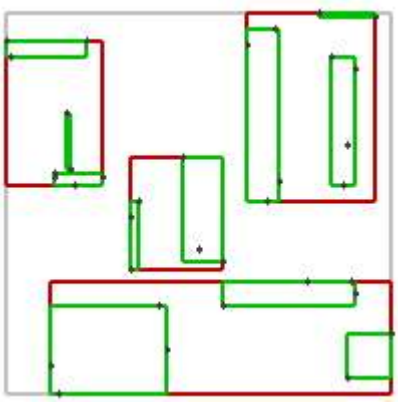
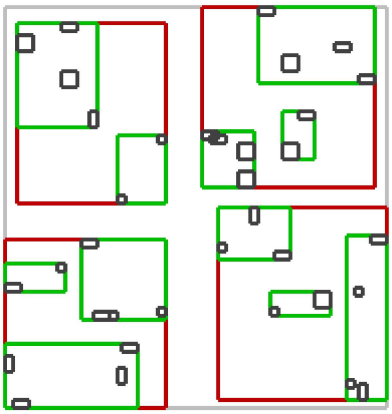
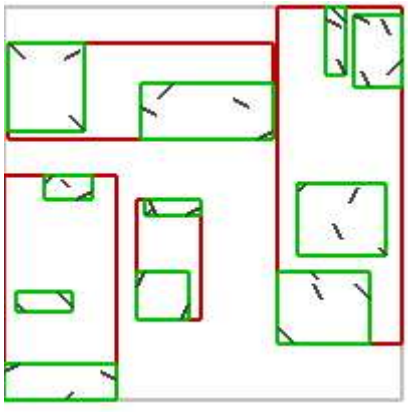
R-tree has 5 parameters but only 2 are required:

```
rtree<Value,
      Parameters,
      IndexableGetter = index::indexable<Value>,
      EqualTo = index::equal_to<Value>,
      Allocator = std::allocator<Value> >
```

- `Value` - type of object which will be stored in the container,
- `Parameters` - parameters type, inserting/splitting algorithm,
- `IndexableGetter` - function object translating `Value` to `Indexable` (`Point` or `Box`) which R-tree can handle,
- `EqualTo` - function object comparing `Values`,
- `Allocator` - `Values` allocator, all allocators needed by the container are created from it.

## *Values and Indexables*

R-tree may store `Values` of any type as long as passed function objects know how to interpret those `Values`, that is extract an `Indexable` that the R-tree can handle and compare `Values`. The `Indexable` is a type adapted to Point, Box or Segment concept. The examples of rtrees storing `Values` translatable to various `Indexables` are presented below.

| rtree<Point, ...> | rtree<Box, ...> | rtree<Segment, ...> |
|---|---|---|
|  |  |  |

By default function objects `index::indexable<Value>` and `index::equal_to<Value>` are defined for some typically used `Value` types which may be stored without defining any additional classes. By default the rtree may store pure `Indexables`, pairs and tuples. In the case of those two collection types, the `Indexable` must be the first stored type.

- `Indexable = Point | Box | Segment`
- `Value = Indexable | std::pair<Indexable, T> | boost::tuple<Indexable, ...> [ | std::tuple<Indexable, ...> ]`

By default `boost::tuple<...>` is supported on all compilers. If the compiler supports C++11 tuples and variadic templates then `std::tuple<...>` may be used "out of the box" as well.

Examples of default `Value` types:

```
geometry::model::point<...>
geometry::model::point_xy<...>
geometry::model::box<...>
geometry::model::segment<...>
std::pair<geometry::model::box<...>, unsigned>
boost::tuple<geometry::model::point<...>, int, float>
```

The predefined `index::indexable<Value>` returns const reference to the `Indexable` stored in the `Value`.

**Important**
The translation is done quite frequently inside the container - each time the rtree needs it.

The predefined `index::equal_to<Value>`:

- for `Point`, `Box` and `Segment` - compares `Values` with geometry::equals().
- for `std::pair<...>` - compares both components of the `Value`. The first value stored in the pair is compared before the second one. If the value stored in the pair is a Geometry, `geometry::equals()` is used. For other types it uses `operator==()`.
- for `tuple<...>` - compares all components of the `Value`. If the component is a `Geometry`, `geometry::equals()` function is used. For other types it uses `operator==()`.

## *Balancing algorithms compile-time parameters*

`Values` may be inserted to the R-tree in many various ways. Final internal structure of the R-tree depends on algorithms used in the insertion process and parameters. The most important is nodes' balancing algorithm. Currently, three well-known types of R-trees may be created.

Linear - classic R-tree using balancing algorithm of linear complexity

```
index::rtree< Value, index::linear<16> > rt;
```

Quadratic - classic R-tree using balancing algorithm of quadratic complexity

```
index::rtree< Value, index::quadratic<16> > rt;
```

R*-tree - balancing algorithm minimizing nodes' overlap with forced reinsertions

```
index::rtree< Value, index::rstar<16> > rt;
```

## *Balancing algorithms run-time parameters*

Balancing algorithm parameters may be passed to the R-tree in run-time. To use run-time versions of the R-tree one may pass parameters which names start with `dynamic_`.

```
// linear
index::rtree<Value, index::dynamic_linear> rt(index::dynamic_linear(16));

// quadratic
index::rtree<Value, index::dynamic_quadratic> rt(index::dynamic_quadratic(16));

// rstar
index::rtree<Value, index::dynamic_rstar> rt(index::dynamic_rstar(16));
```

The obvious drawback is a slightly slower R-tree.

### *Non-default parameters*

Non-default R-tree parameters are described in the reference.

### *Copying, moving and swapping*

The R-tree is copyable and movable container. Move semantics is implemented using Boost.Move library so it's possible to move the container on a compilers without rvalue references support.

```
// default constructor
index::rtree< Value, index::rstar<8> > rt1;

// copy constructor
index::rtree< Value, index::rstar<8> > rt2(r1);

// copy assignment
rt2 = r1;

// move constructor
index::rtree< Value, index::rstar<8> > rt3(boost::move(rt1));

// move assignment
rt3 = boost::move(rt2);

// swap
rt3.swap(rt2);
```

### *Inserting and removing Values*

The following code creates an R-tree using quadratic balancing algorithm.

```
using namespace boost::geometry;
typedef std::pair<Box, int> Value;
index::rtree< Value, index::quadratic<16> > rt;
```

To insert or remove a `Value` by method call one may use the following code.

```
Value v = std::make_pair(Box(...), 0);

rt.insert(v);

rt.remove(v);
```

To insert or remove a `Value` by function call one may use the following code.

```
Value v = std::make_pair(Box(...), 0);

index::insert(rt, v);

index::remove(rt, v);
```

Typically you will perform those operations in a loop in order to e.g. insert some number of `Values` corresponding to geometrical objects (e.g. `Polygons`) stored in another container.

### *Additional interface*

The R-tree allows creation, inserting and removing of Values from a range. The range may be passed as `[first, last)` Iterators pair or as a Range adapted to one of the Boost.Range Concepts.

```
namespace bgi = boost::geometry::index;
typedef std::pair<Box, int> Value;
typedef bgi::rtree< Value, bgi::linear<32> > RTree;

std::vector<Value> values;
/* vector filling code, here */

// create R-tree with default constructor and insert values with insert(Value const&)
RTree rt1;
BOOST_FOREACH(Value const& v, values)
   rt1.insert(v);

// create R-tree with default constructor and insert values with insert(Iter, Iter)
RTree rt2;
rt2.insert(values.begin(), values.end());

// create R-tree with default constructor and insert values with insert(Range)
RTree rt3;
rt3.insert(values_range);

// create R-tree with constructor taking Iterators
RTree rt4(values.begin(), values.end());

// create R-tree with constructor taking Range
RTree rt5(values_range);

// remove values with remove(Value const&)
BOOST_FOREACH(Value const& v, values)
   rt1.remove(v);

// remove values with remove(Iter, Iter)
rt2.remove(values.begin(), values.end());

// remove values with remove(Range)
rt3.remove(values_range);
```

Furthermore, it's possible to pass a Range adapted by one of the Boost.Range adaptors into the rtree (more complete example can be found in the **Examples** section).

```
// create Rtree containing `std::pair<Box, int>` from a container of Boxes on the fly.
RTree rt6(boxes | boost::adaptors::indexed()
                | boost::adaptors::transformed(pair_maker()));
```

### *Insert iterator*

There are functions like `std::copy()`, or R-tree's queries that copy values to an output iterator. In order to insert values to a container in this kind of function insert iterators may be used. Geometry.Index provide its own `bgi::insert_iterator<Container>` which is generated by `bgi::inserter()` function.

```
namespace bgi = boost::geometry::index;
typedef std::pair<Box, int> Value;
typedef bgi::rtree< Value, bgi::linear<32> > RTree;

std::vector<Value> values;
/* vector filling code, here */

// create R-tree and insert values from the vector
RTree rt1;
std::copy(values.begin(), values.end(), bgi::inserter(rt1));

// create R-tree and insert values returned by a query
RTree rt2;
rt1.spatial_query(Box(/*...*/), bgi::inserter(rt2));
```

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